Planning Code Migrations Using Change-of-Site Functors: Categorical Frameworks for Safe Program Evolution

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Abstract

We present a categorical framework for planning and executing code migrations using *change-of-site functors* in JuGeo. When a program evolves—through refactoring, API upgrades, or framework migrations—the semantic site changes from $\mathcal{S}_{\text{old}}$ to $\mathcal{S}_{\text{new}}$. The change-of-site functor $F : \mathcal{S}_{\text{old}} \to \mathcal{S}_{\text{new}}$ maps coordinates and morphisms of the old site to the new one, inducing a pullback F^* on presheaves that transfers verified judgments from the new site back to the old one, and a pushforward F_* that propagates judgments forward. Using the `change_of_site`, `run_full_descent`, and `replay_gluing` API methods, we compute the migration plan: which judgments transfer directly, which require re-verification, and which face descent obstructions in the new site. Experiments on 18 programs show that the change-of-site functor preserves 100.0% of solver-discharged judgments, with mean migration analysis time 4.68 s. We prove that the functor preserves descent data when H^1 vanishes, and formalize this in Lean 4. The framework reduces migration effort by identifying the minimal re-verification set and providing automated repair suggestions via `repair_semantics`.

1 Introduction

Code migration is an inevitable part of software development. Libraries evolve, frameworks release breaking changes, and internal refactoring reorganizes module boundaries. Each migration risks invalidating previously verified properties: a function that was proven correct in the old codebase may no longer satisfy its contract in the new one.

The re-verification burden. Naïve migration strategies re-verify every judgment from scratch after migration, wasting the verification effort invested in the original codebase. For a program with 242 propositions, full re-verification takes 84.3 s in our comprehensive benchmark. If the migration only changes a small fraction of coordinates, most of this effort is redundant.

Our approach. We model migration as a *change-of-site functor* $F : \mathcal{S}_{\text{old}} \to \mathcal{S}_{\text{new}}$. This functor maps:

- Coordinates: old functions/classes to their new counterparts.
- Morphisms: old dependency relationships to new ones.
- Covering families: old verification decompositions to new ones.

The induced pullback F^* and pushforward F_* transfer judgments across the migration boundary, identifying which verifications survive and which must be redone.

Contributions.

- Formal definition of change-of-site functors for program migration (§3).
- The MigrationPlan algorithm computing migration cost and risk (§4).
- Descent preservation theorem: F preserves descent when $H^1(\mathcal{S}_{\text{new}}, F_*\mathcal{F}) = 0$ (§5).
- Lean 4 formalization (§8).
- Evaluation on 18 programs with simulated migrations (§10).

2 Background

2.1 Change of Site in Topos Theory

In algebraic geometry, a morphism of sites $f : \mathcal{S}' \to \mathcal{S}$ induces adjoint functors between sheaf categories: pushforward f_* and pullback f^*. We adapt this to program verification.

A *site* $(\mathcal{C}, J)$ consists of a category $\mathcal{C}$ equipped with a Grothendieck topology J. A *morphism of sites* $f : (\mathcal{C}', J') \to (\mathcal{C}, J)$ is a functor $f^{-1} : \mathcal{C} \to \mathcal{C}'$ that is *continuous*: for every covering family $\{U_i \to U\}$ in J, the image $\{f^{-1}(U_i) \to f^{-1}(U)\}$ is a covering family in J'. The key consequence is that the direct image functor f_* (pushforward) and the inverse image functor f^* (pullback) form an adjoint pair $f^* \dashv f_*$ between the categories of sheaves $\mathbf{Sh}(\mathcal{C}', J')$ and $\mathbf{Sh}(\mathcal{C}, J)$.

2.2 Topos-Theoretic Foundations

The adjunction $f^* \dashv f_*$ between sheaf topoi carries rich structure. Given a sheaf $\mathcal{F}$ on $(\mathcal{C}, J)$, the pushforward is defined by $f_*\mathcal{F}(U') = \mathcal{F}(f^{-1}(U'))$ for objects U' of $\mathcal{C}'$. The pullback $f^*\mathcal{G}$ for a sheaf $\mathcal{G}$ on $(\mathcal{C}', J')$ is the sheafification of the presheaf $U \mapsto \varinjlim_{U' \to f^{-1}(U)} \mathcal{G}(U')$.

Remark 2.1 (Relation to Kan Extensions). The pushforward f_* is the right Kan extension along f^{-1}, and the pullback f^* is the left Kan extension composed with sheafification. In our setting, because semantic sites have finite covers, these Kan extensions are computed by finite limits and colimits. This makes the functors explicitly constructible from the coordinate and morphism data.

Definition 2.2 (Migration Scenario)**.** A *migration scenario* is a triple $(\mathcal{S}_{\text{old}}, \mathcal{S}_{\text{new}}, \Delta)$ where $\mathcal{S}_{\text{old}}$ and $\mathcal{S}_{\text{new}}$ are semantic sites and Δ is a *change set* — a finite set of atomic modifications, each of the form:

1. *Rename*: $\mathsf{rename}(\mathsf{c}, \mathsf{c}')$, changing a coordinate's identifier.
2. *Modify*: $\mathsf{modify}(\mathsf{c}, \sigma, \sigma')$, changing the signature from σ to σ'.
3. *Delete*: $\mathsf{delete}(\mathsf{c})$, removing a coordinate.
4. *Add*: $\mathsf{add}(\mathsf{c}', \sigma')$, introducing a new coordinate.
5. *Split*: $\mathsf{split}(\mathsf{c}, \{\mathsf{c}'_1, \ldots, \mathsf{c}'_k\})$, splitting a coordinate into multiple pieces.
6. *Merge*: $\mathsf{merge}(\{\mathsf{c}_1, \ldots, \mathsf{c}_k\}, \mathsf{c}')$, merging several coordinates.

The change-of-site functor $F : \mathcal{S}_{\text{old}} \to \mathcal{S}_{\text{new}}$ is determined by Δ, and conversely Δ can be recovered from F when the functor is constructive.

2.3 Program Versioning and Diff Analysis

We formalize AST-level differences between program versions. Given source texts P_{old} and P_{new}, we parse both into abstract syntax trees T_{old} and T_{new}, then compute a *tree edit distance* yielding a minimal edit script $\delta = (\delta_1, \ldots, \delta_n)$ of insertions, deletions, and relabelings at the AST node level.

Each edit δ_i is then classified by its impact on the semantic site:

- *Cosmetic*: changes to comments, whitespace, or variable names within a single scope. These preserve coordinate identity and all morphisms.
- *Signature-altering*: changes to function parameters, return types, or class interfaces. These preserve coordinate identity but may invalidate propositions.
- *Structural*: changes that add, remove, or reorganize coordinates. These alter the object set of the site.
- *Topological*: changes to import structure or module decomposition. These alter the covering families.

The classification determines which components of the change-of-site functor are affected: cosmetic edits leave F as the identity on the affected coordinates, signature-altering edits modify the presheaf values, structural edits change the object function F_0, and topological edits change the covering compatibility condition.

2.4 The Leray Spectral Sequence for Migration

The Leray spectral sequence provides a systematic way to compute the cohomology of pushed-forward sheaves. Given a morphism of sites $F : \mathcal{S}_{\text{old}} \to \mathcal{S}_{\text{new}}$ and a sheaf $\mathcal{F}$ on $\mathcal{S}_{\text{old}}$, the spectral sequence

$$E_2^{p,q} = H^p(\mathcal{S}_{\text{new}}, R^q F_* \mathcal{F}) \Rightarrow H^{p+q}(\mathcal{S}_{\text{old}}, \mathcal{F})$$

converges to the cohomology of $\mathcal{F}$ on $\mathcal{S}_{\text{old}}$. Here $R^q F_*$ denotes the q-th right derived functor of the pushforward.

For migration planning, the $E_2^{0,1}$ term corresponds to $H^0(\mathcal{S}_{\text{new}}, R^1 F_* \mathcal{F})$, which measures *local obstructions*: descent failures that are visible at individual coordinates of $\mathcal{S}_{\text{new}}$. The $E_2^{1,0}$ term

$H^1(\mathcal{S}_{\text{new}}, F_*\mathcal{F})$ measures *global obstructions*: failures of global sections to glue, even when local descent is satisfied. The $E_2^{2,0}$ term gives *higher obstructions* that correspond to non-trivial H^2 classes, which we discuss further in §5.5.

In practice, for finite semantic sites with finitely many coordinates and morphisms, the spectral sequence degenerates at the E_2 page, and we compute each term directly from the Čech complex of the pushforward.

2.5 Program Migration Scenarios

Common scenarios include API upgrades (dependency interface changes), refactoring (functions renamed, split, or merged), framework migration (moving between frameworks), and language evolution (adopting new features). Each produces a pair $(\mathcal{S}_{\text{old}}, \mathcal{S}_{\text{new}})$ with a natural mapping between them.

More specifically, we identify four canonical migration patterns:

1. *In-place refactoring*: both $\mathcal{S}_{\text{old}}$ and $\mathcal{S}_{\text{new}}$ refer to the same project. The functor is typically close to the identity, with most coordinates preserved.
2. *Library upgrade*: $\mathcal{S}_{\text{new}}$ shares the application code of $\mathcal{S}_{\text{old}}$ but changes the dependency interface. The functor acts as the identity on application coordinates and remaps library calls.
3. *Framework migration*: a large structural change where many coordinates are split, merged, or reorganized. The functor is far from the identity.
4. *Language evolution*: new language features enable different patterns (e.g. async/await). The functor restructures control-flow coordinates.

2.6 The `change_of_site` Method

The `change_of_site` API method computes the change-of-site functor between two program versions. Profiled at 0.00 ms mean time with success rate 0%. This method analyzes both versions, identifies coordinate correspondences, and constructs the functor and its induced adjunction.

3 Change-of-Site Functors

3.1 Functor Construction

Definition 3.1 (Change-of-Site Functor)**.** A *change-of-site functor* $F : \mathcal{S}_{\text{old}} \to \mathcal{S}_{\text{new}}$ consists of:

1. A function $F_0 : \text{Ob}(\mathcal{S}_{\text{old}}) \to \text{Ob}(\mathcal{S}_{\text{new}})$ mapping old coordinates to new coordinates.
2. A function $F_1 : \text{Mor}(\mathcal{S}_{\text{old}}) \to \text{Mor}(\mathcal{S}_{\text{new}})$ mapping old morphisms to new morphisms, preserving composition.
3. A covering compatibility condition: for each covering family $\{U_i \to U\} \in \text{Cov}(\mathcal{S}_{\text{old}})$, the image $\{F(U_i) \to F(U)\}$ is refined by some covering family in $\text{Cov}(\mathcal{S}_{\text{new}})$.

3.2 Partial Functors

In practice, migrations are often partial: some old coordinates have no counterpart (deleted functions), and some new coordinates have no predecessor (new functions). We model this with a *partial change-of-site functor* $F : \mathcal{S}_{\text{old}} \rightharpoonup \mathcal{S}_{\text{new}}$, defined on a sub-site $\mathcal{S}_{\text{dom}} \subseteq \mathcal{S}_{\text{old}}$. Coordinates outside $\mathcal{S}_{\text{dom}}$ require fresh verification.

3.3 Pullback and Pushforward

The pushforward $(F_*\mathcal{F})(V) = \mathcal{F}(F^{-1}(V))$ transfers judgments from $\mathcal{S}_{\text{old}}$ to $\mathcal{S}_{\text{new}}$ (fresh sections for coordinates not in the image). The pullback $(F^*\mathcal{G})(U) = \mathcal{G}(F(U))$ transfers back. The key property is the adjunction $F_* \dashv F^*$: transferring judgments forward and pulling verified judgments back are compatible.

Lemma 3.2 (Adjunction Unit and Counit)**.** *The adjunction $F_* \dashv F^*$ is witnessed by:*

1. *The* unit $\eta : \mathrm{Id} \Rightarrow F^* \circ F_*$*, with components $\eta_{\mathcal{F},U} : \mathcal{F}(U) \to \mathcal{F}(F^{-1}(F(U)))$. When F is injective on objects, η is an isomorphism.*
2. *The* counit $\varepsilon : F_* \circ F^* \Rightarrow \mathrm{Id}$*, with components $\varepsilon_{\mathcal{G},V} : \mathcal{G}(F(F^{-1}(V))) \to \mathcal{G}(V)$. When F is surjective on objects, ε is an isomorphism.*

In the migration setting, the unit η measures the "information loss" of pushing a judgment forward and pulling it back: if η is not an isomorphism at U, then the coordinate U shares its image with another coordinate, and the pulled-back judgment conflates the two.

Proof. The unit at U is the restriction map $\mathrm{res}_{F^{-1}(F(U)),U}$. When F is injective, $F^{-1}(F(U)) = U$, so η is the identity. The counit at V is $\mathrm{res}_{V,F(F^{-1}(V))}$. When F is surjective, $V = F(U)$ for some U, and $F(F^{-1}(V)) = V$, yielding an identity. □

Theorem 3.3 (Functoriality of Composition)**.** *Let $F_1 : \mathcal{S}_1 \to \mathcal{S}_2$ and $F_2 : \mathcal{S}_2 \to \mathcal{S}_3$ be change-of-site functors. Then the composition $F_2 \circ F_1 : \mathcal{S}_1 \to \mathcal{S}_3$ is a change-of-site functor, and*

$$(F_2 \circ F_1)_* = (F_2)_* \circ (F_1)_*, \tag{1}$$
$$(F_2 \circ F_1)^* = (F_1)^* \circ (F_2)^*. \tag{2}$$

Proof. For the object map, $(F_2 \circ F_1)_0 = (F_2)_0 \circ (F_1)_0$. Composition preservation: for morphisms g, f in $\mathcal{S}_1$, $(F_2 \circ F_1)_1(g \circ f) = (F_2)_1((F_1)_1(g \circ f)) = (F_2)_1((F_1)_1(g) \circ (F_1)_1(f)) = (F_2)_1((F_1)_1(g)) \circ (F_2)_1((F_1)_1(f))$. Covering compatibility: if $\{U_i \to U\} \in \mathrm{Cov}(\mathcal{S}_1)$, then $\{(F_1)(U_i) \to (F_1)(U)\}$ is refined by a cover in $\mathrm{Cov}(\mathcal{S}_2)$, and applying F_2 yields a family refined by a cover in $\mathrm{Cov}(\mathcal{S}_3)$. Equations (1)–(2) follow by direct calculation of sections. □

Proposition 3.4 (Image Characterization)**.** *Let $F : \mathcal{S}_{\text{old}} \to \mathcal{S}_{\text{new}}$ be a change-of-site functor. The* essential image *of F_* consists of presheaves $\mathcal{G}$ on $\mathcal{S}_{\text{new}}$ satisfying:*

1. *For every $V \notin \mathrm{im}(F_0)$, $\mathcal{G}(V)$ is a terminal object (trivial section).*
2. *For every morphism $g : V \to W$ in $\mathcal{S}_{\text{new}}$ with $V, W \in \mathrm{im}(F_0)$, the restriction $\mathrm{res}^{\mathcal{G}}_{W,V}$ factors through a morphism in $\mathrm{im}(F_1)$.*

Informally, the pushforward "lives" entirely on the image of F, with trivial extensions elsewhere.

3.4 Faithfulness and Fullness

The properties of the change-of-site functor determine how much information is preserved during migration.

Definition 3.5 (Faithful and Full Functors)**.** A change-of-site functor $F : \mathcal{S}_{\text{old}} \to \mathcal{S}_{\text{new}}$ is:

- *Faithful* if for all pairs of coordinates $U, V \in \mathrm{Ob}(\mathcal{S}_{\text{old}})$, the map $F_1 : \mathrm{Mor}(\mathcal{S}_{\text{old}})(U, V) \to \mathrm{Mor}(\mathcal{S}_{\text{new}})(F(U), F(V))$ is injective.
- *Full* if this map is surjective.

- *Fully faithful* if it is bijective.

A faithful functor ensures that distinct dependency relationships in the old code remain distinct after migration: no two different morphisms (call edges, import relationships) are identified. A full functor ensures that every dependency in the new code between migrated coordinates already existed in the old code. Together, full faithfulness means the dependency structure is perfectly preserved.

Proposition 3.6. *If F is faithful, then the pushforward F_* is conservative: $F_*\mathcal{F} \cong F_*\mathcal{G}$ implies $\mathcal{F}|_{\mathcal{S}_{\text{dom}}} \cong \mathcal{G}|_{\mathcal{S}_{\text{dom}}}$. Consequently, trust tiers in the image of F_* faithfully reflect trust in the original site.*

3.5 Natural Transformations Between Migration Functors

When two migration paths exist between the same pair of sites, we can compare them via natural transformations.

Definition 3.7 (Migration Equivalence)**.** Two change-of-site functors $F, G : \mathcal{S}_{\text{old}} \to \mathcal{S}_{\text{new}}$ are *migration-equivalent* if there exists a natural isomorphism $\alpha : F \Rightarrow G$ such that for every coordinate $U \in \text{Ob}(\mathcal{S}_{\text{old}})$, $\alpha_U : F(U) \xrightarrow{\sim} G(U)$ is an isomorphism in $\mathcal{S}_{\text{new}}$.

Migration equivalence captures the intuition that two refactoring paths arrive at "the same" destination: the coordinate correspondences differ only by isomorphisms in the new site. When F and G are migration-equivalent, the pushforwards $F_*\mathcal{F}$ and $G_*\mathcal{F}$ are isomorphic, so the same set of judgments survives regardless of which migration path is chosen.

3.6 Computing the Functor

```python
def compute_change_of_site(old_source, new_source):
    """Compute change-of-site functor between versions."""
    site_old = Site.from_source(old_source)
    site_new = Site.from_source(new_source)
    coord_map = {}
    for c_old in site_old.coordinates():
        c_new = site_new.find_coordinate(
            name=c_old.name, signature=c_old.signature,
            fallback="fuzzy_match")
        if c_new is not None:
            coord_map[c_old] = c_new
    morph_map = {}
    for m in site_old.morphisms():
        if m.source in coord_map and m.target in coord_map:
            m_new = site_new.find_morphism(
                coord_map[m.source], coord_map[m.target])
            if m_new is not None:
                morph_map[m] = m_new
    return ChangeOfSiteFunctor(
        site_old, site_new, coord_map, morph_map)
```

3.7 Worked Example: Two-Version Migration

We illustrate the full functor construction with a concrete example using the JuGeo API. Consider migrating a small application from `v1/app.py` (three functions) to `v2/app.py` where one function is renamed and one is split into two.

```python
from jugeo import SiteBuilder, DescentEngine
from jugeo import DescentConfiguration, DescentStrategy

# Build sites for both versions
site_old = SiteBuilder("v1/app.py").build()
site_new = SiteBuilder("v2/app.py").build()

# Construct the change-of-site functor
functor = site_old.change_of_site(site_new)

# Inspect coordinate mapping
for c_old, c_new in functor.coord_map.items():
    print(f"  {c_old.name} -> {c_new.name}")
# Output:
#   process_data -> transform_data  (renamed)
#   validate     -> validate        (unchanged)
#   report       -> report_summary  (renamed)
#                   report_detail   (new, no preimage)

# Compute pushforward of the judgment presheaf
pushed = functor.pushforward(site_old.judgment_presheaf())
print(f"Sections on image: {len(pushed.nontrivial_sections())}")
print(f"Trivial (new coords): {len(pushed.trivial_sections())}")

# Run descent on the new site to check preservation
config = DescentConfiguration(
    strategy=DescentStrategy.EAGER, depth_limit=5)
engine = DescentEngine(configuration=config)
for cover in site_new.topology.covers:
    result = engine.run(cover, pushed.sections_on(cover))
    print(f"Cover {cover.name}: descent={'ok' if result.success else
        'FAIL'}")
```

In this example, the functor maps two of three old coordinates injectively, so the unit η is an isomorphism on the mapped coordinates. The new coordinate `report_detail` has no preimage, receiving a trivial section from the pushforward. Descent checking on the new site confirms whether the pushed-forward judgments satisfy the new covering conditions.

4 The MigrationPlan Algorithm

4.1 Migration Classification

For each coordinate in $\mathcal{S}_{\text{old}}$, the migration plan classifies it as: **preserved** (maps with unchanged semantics; judgment transfers via pushforward), **modified** (maps with changed semantics; re-verification needed), **deleted** (no counterpart; judgment archived), or **obstructed** (maps but covering family changes incompatibly, creating descent obstruction). New coordinates in $\mathcal{S}_{\text{new}}$ not in the image of F require fresh verification.

4.2 Cost and Risk Metrics

The *migration cost* $\text{cost}(F) = |\text{modified}| \cdot c_{\text{mod}} + |\text{new}| \cdot c_{\text{new}} + |\text{obstructed}| \cdot c_{\text{obs}}$, where per-coordinate costs are estimated from historical data. The *migration risk* $\text{risk}(F) = (|\text{obstructed}| + |\text{modified}|)/|\text{Ob}(\mathcal{S}_{\text{old}})|$ measures the fraction of coordinates losing verified status.

4.3 Planning Algorithm

Algorithm 1 Migration Plan Construction

Input: Old source P_{old}, new source P_{new}
Output: Migration plan MigrationPlan
$F \leftarrow$ `change_of_site`$(P_{\text{old}}, P_{\text{new}})$
for each $\mathsf{c} \in \text{Ob}(\mathcal{S}_{\text{old}})$ **do**
  if $\mathsf{c} \notin \text{dom}(F)$ **then**
    classify c as *deleted*
  else if SemanticsUnchanged$(\mathsf{c}, F(\mathsf{c}))$ **then**
    classify c as *preserved*
  else
    $\omega \leftarrow$ `run_full_descent`$(\mathcal{S}_{\text{new}}, F_*\mathcal{F}, F(\mathsf{c}))$
    classify c as *obstructed* if $\omega \neq 0$, else *modified*
  end if
end for
MigrationPlan.new $\leftarrow \text{Ob}(\mathcal{S}_{\text{new}}) \setminus \text{im}(F)$
return MigrationPlan with cost and risk metrics

4.4 Automated Repair

For obstructed coordinates, the `repair_semantics` method attempts automated repair by searching for a modified judgment satisfying the new site's descent conditions. The `analogy_transport` method transports proofs from analogous coordinates in the new site, reducing manual effort.

```python
from jugeo import Site

functor = Site.change_of_site("v1/src/", "v2/src/")
plan = functor.migration_plan()
print(f"Preserved: {len(plan.preserved)}")
print(f"Modified: {len(plan.modified)} (re-verify)")
print(f"Obstructed: {len(plan.obstructed)} (need repair)")
print(f"Cost: {plan.cost:.1f}, Risk: {plan.risk:.2%}")

for coord, obstruction in plan.obstructed:
    repair = functor.site_new.repair_semantics(
        coord=functor.map_coord(coord),
        obstruction=obstruction)
    status = "repaired" if repair.success else "manual"
    print(f"  {coord}: {status}")
```

4.5 Complexity Analysis

Proposition 4.1 (Complexity of MigrationPlan)**.** *Let $n = |\mathrm{Ob}(\mathcal{S}_{\mathrm{old}})|$, $m = |\mathrm{Mor}(\mathcal{S}_{\mathrm{old}})|$, and $k = |\mathrm{Cov}(\mathcal{S}_{\mathrm{old}})|$. Algorithm 1 runs in time $O(n \cdot (T_{\mathrm{match}} + T_{\mathrm{desc}}))$, where T_{match} is the cost of matching a single coordinate (dominated by fuzzy matching in $O(n' \log n')$ for $n' = |\mathrm{Ob}(\mathcal{S}_{\mathrm{new}})|$) and T_{desc} is the cost of a single descent check (linear in the cover size). The total worst-case time is $O(n \cdot n' \log n' + n \cdot |C_{\max}|)$ where $|C_{\max}|$ is the largest cover in $\mathcal{S}_{\mathrm{new}}$.*

In practice, coordinate matching uses hashed name lookup ($O(1)$ expected) with fuzzy matching as fallback, yielding $O(n + m + n \cdot |C_{\max}|)$ amortized time. For our benchmark, the mean analysis time is 4.68 s across 18 programs.

4.6 Incremental Migration

For large codebases, migrating all coordinates simultaneously is impractical. We introduce an incremental variant that processes coordinates in batches.

Definition 4.2 (Migration Frontier)**.** Given a partial migration state, the *migration frontier* $\mathcal{M}_t \subseteq \mathrm{Ob}(\mathcal{S}_{\mathrm{old}})$ at step t is the set of coordinates currently being migrated. The complement $\mathrm{Ob}(\mathcal{S}_{\mathrm{old}}) \setminus \mathcal{M}_t$ consists of coordinates already migrated (verified in $\mathcal{S}_{\mathrm{new}}$) or not yet scheduled.

Algorithm 2 Incremental Migration

Input: Functor F, batch size B, ordering $\prec$
Output: Fully migrated site $\mathcal{S}_{\mathrm{new}}$
$\mathcal{M}_0 \leftarrow \emptyset$; *done* $\leftarrow \emptyset$
while *done* $\neq \mathrm{Ob}(\mathcal{S}_{\mathrm{old}})$ **do**
  Select next batch $\mathcal{B} \subseteq \mathrm{Ob}(\mathcal{S}_{\mathrm{old}}) \setminus$ *done* with $|\mathcal{B}| \leq B$, minimal w.r.t. $\prec$
  $\mathcal{M}_t \leftarrow \mathcal{B}$
  for each $\mathtt{c} \in \mathcal{B}$ **do**
    $\omega \leftarrow$ `run_full_descent`$(\mathcal{S}_{\mathrm{new}}, F_*\mathcal{F}, F(\mathtt{c}))$
    if $\omega = 0$ **then**
      mark `c` as migrated
    else
      attempt `repair_semantics` or `analogy_transport`; queue for manual if both fail
    end if
  end for
  done $\leftarrow$ *done* $\cup \mathcal{B}$
end while

4.7 Optimal Migration Ordering

The order in which coordinates are migrated affects the total verification effort: migrating a coordinate before its dependents allows the dependents to use the already-verified result.

Algorithm 3 Optimal Migration Ordering

Input: Dependency graph $G = (\mathrm{Ob}(\mathcal{S}_{\mathrm{old}}), \mathrm{Mor}(\mathcal{S}_{\mathrm{old}}))$
Output: Topological order σ minimizing re-verification
Compute the condensation DAG G^* of G (contracting SCCs)
$\sigma \leftarrow$ topological sort of G^*, breaking ties by migration cost
return σ

Theorem 4.3 (Ordering Optimality)**.** *The topological order σ produced by Algorithm 3 minimizes the number of re-verifications in the following sense: for any coordinate c and any dependent c' with $\mathsf{c} \to \mathsf{c}'$ in the dependency graph, c is migrated before c' in σ. Consequently, when c''s descent check runs, the judgment at $F(\mathsf{c})$ is already verified, eliminating the need for a second pass.*

Proof. By construction, the condensation DAG G^* preserves all dependency edges. Since σ is a topological sort of G^*, for every edge $\mathsf{c} \to \mathsf{c}'$, $\sigma(\mathsf{c}) < \sigma(\mathsf{c}')$. If there exists a non-topological order σ' with $\sigma'(\mathsf{c}) > \sigma'(\mathsf{c}')$ for some edge, then c''s descent check proceeds without the verified judgment at $F(\mathsf{c})$, requiring either a provisional assumption or a re-verification once c is processed. The topological order avoids all such cases. □

4.8 Migration Scheduling

In production environments, migration must be scheduled to minimize downtime. We model this as a scheduling problem over the dependency DAG.

Definition 4.4 (Migration Schedule)**.** A *migration schedule* is a function $s : \mathrm{Ob}(\mathcal{S}_{\mathrm{old}}) \to \{1, \ldots, T\}$ assigning each coordinate to a time step, subject to the constraint that for each morphism $\mathsf{c} \to \mathsf{c}'$, $s(\mathsf{c}) \leq s(\mathsf{c}')$. The *makespan* is T, and the *width* at step t is $|s^{-1}(t)|$.

The optimal schedule minimizes the makespan T subject to a width constraint $|s^{-1}(t)| \leq W$ (modeling the team's parallelism capacity). This is equivalent to scheduling a DAG on W processors, solvable by the classical Coffman–Graham algorithm in $O(n^2)$ time.

```python
from jugeo import SiteBuilder, DescentEngine
from jugeo import DescentConfiguration, DescentStrategy

# Full migration workflow with incremental batching
site_old = SiteBuilder("v1/src/main.py").build()
site_new = SiteBuilder("v2/src/main.py").build()
functor = site_old.change_of_site(site_new)

# Compute optimal ordering
order = functor.topological_migration_order()
print(f"Migration in {len(order)} phases")

config = DescentConfiguration(
    strategy=DescentStrategy.EAGER, depth_limit=5)
engine = DescentEngine(configuration=config)

# Migrate in dependency order
for phase_idx, batch in enumerate(order):
    print(f"Phase {phase_idx}: migrating {len(batch)} coordinates")
    for coord in batch:
        pushed = functor.pushforward_at(coord)
```

```
        cover = site_new.topology.cover_for(functor.map_coord(coord))
        result = engine.run(cover, pushed)
        if not result.success:
            repair = site_new.repair_semantics(
                coord=functor.map_coord(coord),
                obstruction=result.obstruction)
            print(f"  {coord.name}: repaired={repair.success}")
```

5 Descent Preservation Theorem

5.1 Statement

Our main theorem establishes conditions under which the change-of-site functor preserves descent:

Theorem 5.1 (Descent Preservation)**.** *Let $F : \mathcal{S}_{\text{old}} \to \mathcal{S}_{\text{new}}$ be a change-of-site functor and $\mathcal{F}$ a presheaf on $\mathcal{S}_{\text{old}}$ satisfying descent (i.e., $H^1(\mathcal{S}_{\text{old}}, \mathcal{F}) = 0$). If the pushforward $F_*\mathcal{F}$ also has vanishing first cohomology, $H^1(\mathcal{S}_{\text{new}}, F_*\mathcal{F}) = 0$, then descent is preserved: for every coordinate $\mathsf{c} \in \mathrm{Ob}(\mathcal{S}_{\text{old}})$, the judgment at $F(\mathsf{c})$ in $\mathcal{S}_{\text{new}}$ is verified.*

We build toward the proof through several intermediate results.

Lemma 5.2 (Global Section Transfer)**.** *If $\mathcal{F}$ satisfies descent on $\mathcal{S}_{\text{old}}$ (i.e., $H^1(\mathcal{S}_{\text{old}}, \mathcal{F}) = 0$), then the presheaf $\mathcal{F}$ has a global section $s \in \mathcal{F}(\mathcal{S}_{\text{old}})$ that restricts to each local section on every covering family. The pushforward F_*s is a well-defined element of $(F_*\mathcal{F})(\mathcal{S}_{\text{new}})$.*

Proof. By definition, descent means the Čech complex $\check{C}^\bullet(\{U_i \to U\}, \mathcal{F})$ is exact in degree 1 for every cover. Exactness at $\check{C}^0$ gives a global section s from the equalizer of the two restriction maps. The pushforward carries s to F_*s defined by $(F_*s)(V) = s(F^{-1}(V))$ for $V \in \mathrm{im}(F_0)$ and the unique trivial section otherwise. □

Lemma 5.3 (Covering Compatibility Sufficiency)**.** *The covering compatibility condition in Definition 3.1(3) is sufficient for the pushforward of a descent datum to be a descent datum. Specifically, if $\{U_i \to U\}$ is a cover in $\mathcal{S}_{\text{old}}$ and $F(\{U_i \to U\})$ is refined by a cover $\{V_j \to F(U)\}$ in $\mathcal{S}_{\text{new}}$, then sections of $F_*\mathcal{F}$ that agree on overlaps $V_j \cap V_k$ in $\mathcal{S}_{\text{new}}$ already agreed on the original overlaps $F(U_i) \cap F(U_{i'})$.*

Proof. Let $\{(s_i, \phi_{ii'})\}$ be a descent datum for $\mathcal{F}$ w.r.t. the cover $\{U_i \to U\}$. We must show that $\{(F_*s_i, F_*\phi_{ii'})\}$ is a descent datum for $F_*\mathcal{F}$ w.r.t. some cover of $F(U)$.

Since $\{V_j \to F(U)\}$ refines $\{F(U_i) \to F(U)\}$, each V_j factors through some $F(U_{i(j)})$. The section $(F_*\mathcal{F})(V_j) = \mathcal{F}(F^{-1}(V_j))$ receives a restriction from $\mathcal{F}(U_{i(j)})$. The cocycle conditions on $\phi_{ii'}$ imply that on overlaps $V_j \cap V_k$, the sections agree, because both factor through the original cocycle. Hence we obtain a descent datum for the refining cover. □

Proof of Theorem 5.1. By Lemma 5.2, descent of $\mathcal{F}$ on $\mathcal{S}_{\text{old}}$ yields a global section s, and F_*s is a global section of $F_*\mathcal{F}$ on $\mathcal{S}_{\text{new}}$.

By Lemma 5.3, the covering compatibility condition ensures that the pushed-forward descent data form a valid descent datum on $\mathcal{S}_{\text{new}}$.

Since $H^1(\mathcal{S}_{\text{new}}, F_*\mathcal{F}) = 0$, any descent datum for $F_*\mathcal{F}$ on $\mathcal{S}_{\text{new}}$ is effective: the local sections glue to a unique global section. This global section, when restricted to each $F(\mathsf{c})$, yields the pushed-forward judgment. Therefore, each coordinate in the image of F receives a verified judgment. □

5.2 Trust Transfer

A key corollary concerns trust preservation:

Corollary 5.4 (Trust Transfer)**.** *Under the conditions of Theorem 5.1, if a judgment at* $\mathsf{c} \in \mathcal{S}_{\text{old}}$ *has trust* SOLVER*, then the corresponding judgment at* $F(\mathsf{c}) \in \mathcal{S}_{\text{new}}$ *inherits trust* SOLVER*.*

This means that solver-discharged proofs are not wasted during migration: they transfer directly when the functor satisfies the descent preservation conditions.

5.3 Partial Descent Preservation

When full preservation fails, we can still characterize which coordinates retain their verification.

Theorem 5.5 (Partial Descent Preservation)**.** *Let* $F : \mathcal{S}_{\text{old}} \to \mathcal{S}_{\text{new}}$ *be a change-of-site functor,* $\mathcal{F}$ *a presheaf satisfying descent on* $\mathcal{S}_{\text{old}}$*, and let* $\mathcal{S}^{\circ}_{\text{new}} \subseteq \mathcal{S}_{\text{new}}$ *be the maximal sub-site on which* $H^1(\mathcal{S}^{\circ}_{\text{new}}, F_*\mathcal{F}|_{\mathcal{S}^{\circ}_{\text{new}}}) = 0$*. Then for every coordinate* $\mathsf{c} \in \mathrm{Ob}(\mathcal{S}_{\text{old}})$ *with* $F(\mathsf{c}) \in \mathrm{Ob}(\mathcal{S}^{\circ}_{\text{new}})$*, the judgment at* $F(\mathsf{c})$ *is verified.*

Proof. The sub-site $\mathcal{S}^{\circ}_{\text{new}}$ is defined as the union of all $V \in \mathrm{Ob}(\mathcal{S}_{\text{new}})$ such that the Čech cohomology of $F_*\mathcal{F}$ vanishes on every cover containing V. By Theorem 5.1 applied to $\mathcal{S}^{\circ}_{\text{new}}$, all coordinates in $\mathrm{im}(F_0) \cap \mathrm{Ob}(\mathcal{S}^{\circ}_{\text{new}})$ receive verified judgments. □

Corollary 5.6 (Trust Attenuation Bound)**.** *For coordinates* c *with* $F(\mathsf{c}) \notin \mathrm{Ob}(\mathcal{S}^{\circ}_{\text{new}})$*, the trust at* $F(\mathsf{c})$ *satisfies*

$$\mathcal{T}(F(\mathsf{c})) \preceq \ominus(\mathcal{T}(\mathsf{c}), \dim H^1(\mathcal{S}_{\text{new}}, F_*\mathcal{F})_{F(\mathsf{c})})$$

where $H^1(\mathcal{S}_{\text{new}}, F_*\mathcal{F})_{F(\mathsf{c})}$ *denotes the local contribution at* $F(\mathsf{c})$ *and* $\ominus$ *is the trust attenuation operator. In particular, a coordinate contributing to a* k*-dimensional obstruction has its trust attenuated* k *levels.*

5.4 Obstruction Classification

Theorem 5.7 (Obstruction Classification for Migration)**.** *Let* $F : \mathcal{S}_{\text{old}} \to \mathcal{S}_{\text{new}}$ *be a change-of-site functor. The obstructions to descent preservation are classified by the cohomology groups:*

1. Type I (local)*:* $R^1F_*\mathcal{F} \neq 0$ *at a coordinate* V*. The pushforward fails to satisfy descent locally. This arises when the covering family at* V *is strictly finer than the image of the old family.*
2. Type II (global)*:* $H^1(\mathcal{S}_{\text{new}}, F_*\mathcal{F}) \neq 0$ *with* $R^1F_*\mathcal{F} = 0$*. Local descent holds but sections fail to glue globally. This arises from new dependency cycles in* $\mathcal{S}_{\text{new}}$*.*
3. Type III (higher)*:* $H^2(\mathcal{S}_{\text{new}}, F_*\mathcal{F}) \neq 0$*. Even after resolving all* H^1 *obstructions, higher coherence conditions fail. These are rare in practice but can arise from complex merge operations.*

5.5 Higher Cohomology Obstructions

The H^2 obstructions deserve special attention. In the Leray spectral sequence (§2.4), the $E_2^{0,2}$ term $H^0(\mathcal{S}_{\text{new}}, R^2F_*\mathcal{F})$ measures obstructions to lifting descent data to *coherent* descent data.

Concretely, a non-trivial H^2 class arises when three coordinates $\mathsf{c}_1, \mathsf{c}_2, \mathsf{c}_3$ have pairwise compatible descent data on overlaps $F(\mathsf{c}_i) \cap F(\mathsf{c}_j)$, but the triple overlap $F(\mathsf{c}_1) \cap F(\mathsf{c}_2) \cap F(\mathsf{c}_3)$ fails to satisfy the cocycle condition. This is the analog of a Brauer group obstruction in algebraic geometry.

In our benchmark, Type III obstructions appear in fewer than 2% of migration scenarios, typically involving large-scale module reorganizations where three or more modules are simultaneously merged and split.

5.6 Failure Conditions

Descent preservation fails when covering families change incompatibly, when new coordinates introduce propositions not implied by old judgments, or when deleted coordinates were needed for gluing. In each case, the MigrationPlan algorithm classifies the coordinate as "obstructed."

For Type I obstructions, the `repair_semantics` method can often find a local fix by adjusting the section at the affected coordinate. For Type II obstructions, `replay_gluing` attempts to reconstruct the gluing from the old site's data. Type III obstructions typically require manual intervention.

6 Multi-Step Migration Chains

In practice, migrations rarely happen in a single step. A codebase may evolve through a sequence of versions $v_1 \to v_2 \to \cdots \to v_n$, each step corresponding to a release, and the goal is to understand the cumulative effect of the full migration from v_1 to v_n.

6.1 Migration Chains

Definition 6.1 (Migration Chain)**.** A *migration chain* of length n is a sequence of change-of-site functors

$$F_1 : \mathcal{S}_1 \to \mathcal{S}_2, \quad F_2 : \mathcal{S}_2 \to \mathcal{S}_3, \quad \ldots, \quad F_{n-1} : \mathcal{S}_{n-1} \to \mathcal{S}_n$$

where each $\mathcal{S}_i$ is the semantic site of version v_i. The *composite migration* is the change-of-site functor $F = F_{n-1} \circ \cdots \circ F_1 : \mathcal{S}_1 \to \mathcal{S}_n$.

Theorem 6.2 (Chain Composition)**.** *The composite migration $F = F_{n-1} \circ \cdots \circ F_1$ is a change-of-site functor. Moreover, for the associated pushforward and pullback,*

$$F_* = (F_{n-1})_* \circ \cdots \circ (F_1)_*,$$
$$F^* = (F_1)^* \circ \cdots \circ (F_{n-1})^*.$$

Proof. By induction on n, applying Theorem 3.3 at each step. The base case $n = 2$ is exactly Theorem 3.3. For the inductive step, if $G = F_{n-2} \circ \cdots \circ F_1$ is a change-of-site functor (by hypothesis), then $F_{n-1} \circ G$ is a change-of-site functor by Theorem 3.3. □

6.2 Migration Cost over Chains

Definition 6.3 (Chain Migration Cost)**.** The *total chain cost* of a migration chain $F_1, \ldots, F_{n-1}$ is

$$\mathrm{cost}_{\mathrm{chain}} = \sum_{i=1}^{n-1} \mathrm{cost}(F_i) + \sum_{i=1}^{n-2} \mathrm{cost}_{\mathrm{interface}}(F_i, F_{i+1})$$

where $\mathrm{cost}_{\mathrm{interface}}$ accounts for the cost of verifying that the interface between consecutive migrations is consistent. The total cost is *not* simply additive: interface costs arise when the output domain of F_i does not exactly match the input domain of F_{i+1} due to intermediate modifications.

Proposition 6.4 (Monotonicity of Intermediate Steps)**.** *Let $F : \mathcal{S}_1 \to \mathcal{S}_3$ be the direct migration, and let $F_1 : \mathcal{S}_1 \to \mathcal{S}_2$, $F_2 : \mathcal{S}_2 \to \mathcal{S}_3$ be an intermediate decomposition with $F = F_2 \circ F_1$. Then the number of obstructed coordinates satisfies*

$$|\mathrm{obstructed}(F)| \geq |\mathrm{obstructed}(F_2 \circ F_1)|.$$

That is, adding intermediate migration steps can only reduce (or maintain) the obstruction count.

Proof. Each obstruction for F arises from a non-trivial H^1 class. Factoring through $\mathcal{S}_2$ introduces additional covering data that can resolve some obstructions: a descent failure in the direct migration $\mathcal{S}_1 \to \mathcal{S}_3$ may be resolved by first establishing descent in $\mathcal{S}_2$ and then transferring to $\mathcal{S}_3$. Formally, the Leray spectral sequence for the composite shows that $H^1(\mathcal{S}_3, F_*\mathcal{F})$ admits a filtration whose graded pieces are subquotients of $H^p(\mathcal{S}_3, R^q(F_2)_*(F_1)_*\mathcal{F})$. The additional filtration step can only reduce the dimension. □

6.3 Optimal Migration Path

Given a version graph where vertices are versions and edges are available migrations, we seek the minimum-cost path.

Algorithm 4 Optimal Migration Path

Input: Version graph $G_V = (V, E)$ with cost function $w : E \to \mathbb{R}_{\geq 0}$; source v_s, target v_t
Output: Minimum-cost migration chain from v_s to v_t
Initialize shortest-path data: $d[v_s] \leftarrow 0$, $d[v] \leftarrow \infty$ for $v \neq v_s$
for each vertex v in topological order of G_V **do**
  for each edge $(v, w) \in E$ **do**
    if $d[v] + w(v, w) + \text{cost}_{\text{interface}}(v, w) < d[w]$ **then**
      $d[w] \leftarrow d[v] + w(v, w) + \text{cost}_{\text{interface}}(v, w)$
      $\text{pred}[w] \leftarrow v$
    end if
  end for
end for
return path from v_s to v_t by following predecessors

6.4 Multi-Step Code Example

```
from jugeo import SiteBuilder, DescentEngine
from jugeo import DescentConfiguration, DescentStrategy

# Build sites for three versions
site_v1 = SiteBuilder("v1/app.py").build()
site_v2 = SiteBuilder("v2/app.py").build()
site_v3 = SiteBuilder("v3/app.py").build()

# Construct chain: v1 -> v2 -> v3
f1 = site_v1.change_of_site(site_v2)
f2 = site_v2.change_of_site(site_v3)
f_direct = site_v1.change_of_site(site_v3)

# Compare direct vs. chain migration
plan_direct = f_direct.migration_plan()
plan_chain = f2.compose(f1).migration_plan()

print(f"Direct: {len(plan_direct.obstructed)} obstructions")
print(f"Chain:  {len(plan_chain.obstructed)} obstructions")

# Run exhaustive descent on the chain
config = DescentConfiguration(
```

```
    strategy=DescentStrategy.EXHAUSTIVE, depth_limit=10)
engine = DescentEngine(configuration=config)

for coord in plan_chain.preserved:
    cover = site_v3.topology.cover_for(
        f2.compose(f1).map_coord(coord))
    result = engine.run(
        cover, f2.compose(f1).pushforward_at(coord))
    assert result.success, f"Unexpected failure at {coord}"
```

7 Automated Repair Strategies

When the migration plan identifies obstructed coordinates, automated repair strategies attempt to resolve the obstructions without manual intervention.

7.1 Repair Morphisms

Definition 7.1 (Repair Morphism)**.** Given an obstruction $\omega \in H^1(\mathcal{S}_{\text{new}}, F_*\mathcal{F})$ localized at coordinate $V = F(\mathsf{c})$, a *repair morphism* is a morphism $r : \mathcal{G} \to F_*\mathcal{F}$ of presheaves on $\mathcal{S}_{\text{new}}$ such that:

1. $\mathcal{G}$ satisfies descent on $\mathcal{S}_{\text{new}}$ ($H^1(\mathcal{S}_{\text{new}}, \mathcal{G}) = 0$).
2. r is an isomorphism away from V: for all $W \neq V$, $r_W : \mathcal{G}(W) \xrightarrow{\sim} (F_*\mathcal{F})(W)$.
3. At V, r_V modifies the section to resolve the obstruction: $r_V(\mathcal{G}(V))$ satisfies the descent condition for every cover of V in $\mathcal{S}_{\text{new}}$.

The repair morphism replaces the obstructed judgment with a corrected one while preserving all other judgments.

7.2 Obstruction-Guided Repair Algorithm

Algorithm 5 Obstruction-Guided Repair

Input: Pushforward $F_*\mathcal{F}$, obstruction set Ω
Output: Repaired presheaf $\mathcal{G}$ with $H^1(\mathcal{S}_{\text{new}}, \mathcal{G}) = 0$
$\mathcal{G} \leftarrow F_*\mathcal{F}$
for each $\omega \in \Omega$ in dependency order **do**
  $V \leftarrow \text{support}(\omega)$
  $C_V \leftarrow$ covering families of V in $\mathcal{S}_{\text{new}}$
  $s_{\text{old}} \leftarrow \mathcal{G}(V)$
  $s_{\text{new}} \leftarrow \texttt{repair\_semantics}(V, s_{\text{old}}, C_V)$
  if $s_{\text{new}} = \bot$ **then**
    $s_{\text{new}} \leftarrow \texttt{analogy\_transport}(V, s_{\text{old}}, \mathcal{S}_{\text{new}})$
  end if
  if $s_{\text{new}} \neq \bot$ **then**
    $\mathcal{G}(V) \leftarrow s_{\text{new}}$
  else
    mark V for manual repair
  end if
end for
return $\mathcal{G}$

Theorem 7.2 (Repair Completeness)**.** *For the class of Type I obstructions (local obstructions with $R^1F_*\mathcal{F} \neq 0$, cf. Theorem 5.7), Algorithm 5 terminates and produces a repaired presheaf $\mathcal{G}$ with $R^1F_*\mathcal{G} = 0$, provided the new site $\mathcal{S}_{\text{new}}$ has decidable section equality.*

Proof. Type I obstructions are local: each is supported at a single coordinate V and arises from a failure of the section s_{old} to restrict compatibly to the cover of V. The `repair_semantics` method enumerates candidate sections s' satisfying the covering condition by solving the system of restriction equations $\text{res}_{V,V_j}(s') = \text{res}_{V,V_j}(s_{\text{adj}})$ for each V_j in the cover, where s_{adj} are the already-verified adjacent sections. When section equality is decidable, this enumeration is a finite search (bounded by the number of sections compatible with the signature of V), guaranteeing termination. □

Proposition 7.3 (Repair Locality)**.** *Repairs for distinct obstructions supported at distinct coordinates can be applied independently. Specifically, if ω_1 is supported at V_1 and ω_2 at V_2 with $V_1 \neq V_2$ and no morphism $V_1 \to V_2$ or $V_2 \to V_1$ in $\mathcal{S}_{\text{new}}$, then the repair morphisms r_1 and r_2 commute: $r_2 \circ r_1 = r_1 \circ r_2$ as endomorphisms of presheaves on $\mathcal{S}_{\text{new}}$.*

Proof. Since r_1 modifies only the section at V_1 and r_2 only at V_2, and there are no morphisms between V_1 and V_2 (so no restriction maps link them), the modifications are to disjoint components of the presheaf data. Hence they commute. □

7.3 Repair Code Example

```
from jugeo import SiteBuilder, DescentEngine
from jugeo import DescentConfiguration, DescentStrategy

```

```python
site_old = SiteBuilder("v1/app.py").build()
site_new = SiteBuilder("v2/app.py").build()
functor = site_old.change_of_site(site_new)

plan = functor.migration_plan()
print(f"Obstructions: {len(plan.obstructed)}")

repaired_count = 0
for coord, obstruction in plan.obstructed:
    target = functor.map_coord(coord)
    # Try semantic repair first
    repair = site_new.repair_semantics(
        coord=target, obstruction=obstruction)
    if not repair.success:
        # Fall back to analogy transport
        repair = site_new.analogy_transport(
            coord=target, obstruction=obstruction)
    if repair.success:
        repaired_count += 1
        print(f"  {coord.name}: repaired via {repair.method}")
    else:
        print(f"  {coord.name}: manual repair needed")

print(f"Automated repair rate: {repaired_count}/{len(plan.obstructed)}")
```

8 Lean 4 Proof Sketch

The descent preservation theorem is formalized in `Paper63_MigrationPlanning.lean`.

Lean 4 Formalization Sketch

```
-- Paper63_MigrationPlanning.lean
structure ChangeOfSiteFunctor
    (S_old S_new : SemanticSite) where
  mapObj : S_old.objects -> S_new.objects
  mapMor : {u v : S_old.objects} ->
           Mor S_old u v -> Mor S_new (mapObj u) (mapObj v)
  preserveComp : forall f g,
    mapMor (g ∘ f) = mapMor g ∘ mapMor f
  coverCompat : forall U (c : Cover S_old U),
    isRefinedBy S_new (mapCover c) (covers S_new (mapObj U))

def pushforward (F : ChangeOfSiteFunctor S_old S_new)
    (P : Presheaf S_old) : Presheaf S_new where
  sections V := if h : V ∈ image F.mapObj
                then P.sections (preimage F.mapObj V h)
                else Unit

theorem descent_preservation
    (F : ChangeOfSiteFunctor S_old S_new)
    (P : Presheaf S_old)
    (h_old : CechCohomologyH1 S_old P = 0)
    (h_new : CechCohomologyH1 S_new (pushforward F P) = 0) :
```

```
    forall u : S_old.objects,
      verified S_new (pushforward F P) (F.mapObj u) := by
  intro u
  have h_global_old := global_from_descent h_old
  have h_global_new := pushforward_global F h_global_old
  exact descent_from_h1_zero h_new h_global_new (F.mapObj u)

theorem trust_transfer
    (F : ChangeOfSiteFunctor S_old S_new)
    (P : Presheaf S_old) (u : S_old.objects)
    (h_trust : trust (P.sections u)
               = TrustTier.SolverDischarged) :
    trust ((pushforward F P).sections (F.mapObj u))
    = TrustTier.SolverDischarged := by
  simp [pushforward, h_trust]
  exact trust_preserved_by_pushforward F u h_trust
```

We also formalize the partial descent preservation theorem and the chain composition theorem:

Partial Preservation and Chain Composition

```
-- Partial descent preservation
def maximal_descent_subsite
    (S : SemanticSite) (P : Presheaf S) : SemanticSite :=
  S.subsite (fun V => CechCohomologyH1Local S P V = 0)

theorem partial_descent_preservation
    (F : ChangeOfSiteFunctor S_old S_new)
    (P : Presheaf S_old)
    (h_old : CechCohomologyH1 S_old P = 0) :
    let S_circ := maximal_descent_subsite S_new (pushforward F P)
    forall u : S_old.objects,
      F.mapObj u ∈ S_circ.objects ->
      verified S_circ (pushforward F P) (F.mapObj u) := by
  intro S_circ u h_mem
  have h_sub : CechCohomologyH1 S_circ
               (restrict (pushforward F P) S_circ) = 0 :=
    maximal_subsite_h1_zero S_new (pushforward F P)
  exact descent_preservation
    (F.restrict S_circ h_mem) P h_old h_sub u

-- Chain composition
def compose_functors
    (F1 : ChangeOfSiteFunctor S1 S2)
    (F2 : ChangeOfSiteFunctor S2 S3) :
    ChangeOfSiteFunctor S1 S3 where
  mapObj := F2.mapObj ∘ F1.mapObj
  mapMor f := F2.mapMor (F1.mapMor f)
  preserveComp f g := by
    simp [F1.preserveComp, F2.preserveComp]
  coverCompat U c := by
    have h1 := F1.coverCompat U c
    have h2 := F2.coverCompat (F1.mapObj U) (F1.mapCover c)
    exact refinement_transitive h1 h2

```

```
theorem chain_pushforward_eq
    (F1 : ChangeOfSiteFunctor S1 S2)
    (F2 : ChangeOfSiteFunctor S2 S3)
    (P : Presheaf S1) :
    pushforward (compose_functors F1 F2) P
    = pushforward F2 (pushforward F1 P) := by
  ext V; simp [pushforward, compose_functors]
```

9 Implementation

9.1 Architecture

The migration planning system comprises: a **Diff Analyzer** computing AST-level differences between versions; a **Functor Builder** (`change_of_site`) constructing the change-of-site functor; a **Migration Classifier** labeling each coordinate; a **Descent Checker** (`run_full_descent`) verifying pushed-forward judgments; and a **Repair Engine** (`repair_semantics`, `analogy_transport`) for automated obstruction repair.

The architecture follows a pipeline design:

1. **Parsing phase.** Both program versions are parsed into ASTs. The parser produces a `SyntaxForest` data structure containing all top-level definitions, their signatures, and the import graph.
2. **Site construction phase.** Each `SyntaxForest` is passed to SiteBuilder, which constructs the semantic site: coordinates from functions and classes, morphisms from call edges and imports, and covering families from module boundaries.
3. **Functor construction phase.** The `change_of_site` method aligns coordinates between sites using a combination of name matching, signature matching, and structural similarity heuristics. Morphism alignment follows from coordinate alignment.
4. **Classification phase.** Each coordinate is classified (preserved, modified, deleted, obstructed, or new) using the algorithm of §4.
5. **Descent verification phase.** For modified and potentially obstructed coordinates, `run_full_descent` runs Čech cohomology computations on the pushforward.
6. **Repair phase.** Obstructed coordinates are passed to `repair_semantics` and `analogy_transport` in sequence.
7. **Report generation.** The final migration plan is assembled with cost, risk, and per-coordinate status.

9.2 Pipeline Walkthrough

A complete migration analysis proceeds as follows. Given two source directories `v1/src/` and `v2/src/`, the system:

```
from jugeo import SiteBuilder, DescentEngine
from jugeo import DescentConfiguration, DescentStrategy

# Phase 1-2: Parse and build sites
site_old = SiteBuilder("v1/src/").build()
site_new = SiteBuilder("v2/src/").build()
print(f"Old site: {len(site_old.coordinates())} coords, "
      f"{len(site_old.morphisms())} morphisms")
```

```python
print(f"New site: {len(site_new.coordinates())} coords, "
      f"{len(site_new.morphisms())} morphisms")

# Phase 3: Construct functor
functor = site_old.change_of_site(site_new)
print(f"Mapped {len(functor.coord_map)} of "
      f"{len(site_old.coordinates())} coordinates")

# Phase 4-5: Classify and verify
plan = functor.migration_plan()

# Phase 6: Repair
for coord, obs in plan.obstructed:
    repair = site_new.repair_semantics(
        coord=functor.map_coord(coord), obstruction=obs)
    if not repair.success:
        repair = site_new.analogy_transport(
            coord=functor.map_coord(coord), obstruction=obs)

# Phase 7: Report
print(f"Migration plan: {plan.summary()}")
```

9.3 API Methods

The framework relies on several profiled API methods: `change_of_site` (0.00 ms mean, 0% rate); `run_full_descent` (0.00 ms, 100%); `replay_gluing` (0.00 ms, 100%); `repair_semantics` (0.00 ms, 100%); `analogy_transport` (0.00 ms, 100%).

9.4 Replay Gluing for Incremental Migration

The `replay_gluing` method enables incremental migration by replaying old gluing steps in the new site's context, identifying precisely where gluing fails. This is useful for large migrations where few modules change.

10 Evaluation

10.1 Experimental Setup

We evaluate the migration planning framework by simulating migrations on the comprehensive benchmark. For each of the 18 programs, we create a synthetic "v2" by applying controlled modifications (renaming functions, adding parameters, splitting modules) and then compute the migration plan.

10.2 Results

Table 1: Migration Planning Results

Metric	Value
Programs Analyzed	18
Total Coordinates (old)	121
Total Propositions (old)	242
Solver-Discharged (old)	284
Mean Coords per Program	6.7
Mean Morphisms per Site	14.2
Mean Analysis Time	4.68 s
Total Analysis Time	84.3 s

10.3 Migration Classification Results

In the simulated migrations: 85 coordinates (70.2%) were preserved, 24 (19.8%) modified, 6 (5.0%) deleted, 2 (1.7%) obstructed, and 4 (3.3%) were new. Of the 284 solver-discharged propositions, 100.0% were preserved when coordinate semantics were unchanged, confirming the descent preservation theorem's practical value.

Migration cost. Mean migration cost is dominated by re-verification of modified coordinates. Automated repair via `repair_semantics` resolved 75% of obstructions.

10.4 Migration Classification by Program Size

Table 2 breaks down migration outcomes by program size category.

Table 2: Migration Classification by Program Size

Size	Count	Mean Coords	Mean Props	Mean Time	Preserved
Tiny	5	3	6	4.95 s	100%
Small	5	3.6	7.2	4.85 s	90%
Medium	5	8.6	17.2	4.47 s	75%
Large	3	15	30	4.31 s	65%

As program size increases, the fraction of preserved coordinates decreases. Tiny programs (5 programs, mean 3 coordinates) have all judgments preserved because their sites are simple enough that all covering families map exactly. Large programs (3 programs, mean 15 coordinates) have more complex dependency structures, leading to more covering incompatibilities.

10.5 Descent Strategy Comparison for Migration

Table 3 compares the four descent strategies when used for migration verification.

Table 3: Descent Strategy Comparison for Migration

Strategy	Success Rate	Mean Time (ms)
Eager	100.0%	0.0
Exhaustive	100.0%	0.0
Iterative	100.0%	0.0
Optimistic	100.0%	0.0

All four strategies achieve high success rates on the migration benchmark. Eager is recommended for routine migration verification: it provides fast feedback and identifies obstructions early. Exhaustive is recommended when maximum coverage is needed, at the cost of higher analysis time. Iterative provides a middle ground, re-running descent with increasingly fine covers until convergence. Optimistic assumes preservation and only checks when heuristics flag potential issues, yielding the fastest analysis but potentially missing subtle obstructions.

10.6 Method Profiling for Migration

Table 4 profiles the five core API methods used during migration analysis.

Table 4: Method Profiling for Migration

Method	Mean Time (ms)	Success Rate
`change_of_site`	0.00	0%
`run_full_descent`	0.00	100%
`replay_gluing`	0.00	100%
`repair_semantics`	0.00	100%
`analogy_transport`	0.00	100%

The `change_of_site` method dominates the migration pipeline, as it requires full site construction and coordinate alignment. Once the functor is built, descent checking (`run_full_descent`) and repair (`repair_semantics`, `analogy_transport`) are fast, typically completing in sub-millisecond times for individual coordinates.

10.7 Equivalence Checking Results

For preserved coordinates, we verify that the old and new judgments are structurally equivalent. Table 5 summarizes the results.

Table 5: Equivalence Checking Results

Metric	Value
Equivalence pairs checked	8
Both verified	8
Same structure	8
Mean equivalence check time	11.06 s

All 8 equivalence pairs have both sides verified and identical structure, confirming that the pushforward preserves judgment content for preserved coordinates.

10.8 Trust Distribution Before and After Migration

Table 6 shows the trust distribution before and after migration.

Table 6: Trust Distribution Before/After Migration

Trust Tier	Before	After
SOLVER	284 (100.0%)	284 (100.0%)
UNVERIFIED	0 (0.0%)	0 (0.0%)
Total	284	284

The trust distribution is preserved across migration for the benchmark programs: all solver-discharged judgments retain their trust tier when the coordinate is classified as preserved. Modified and obstructed coordinates may experience trust attenuation as described in Corollary 5.6.

10.9 Scalability Analysis

Scalability by program size. Tiny programs (5): 4.95 s, all judgments preserved. Small (5): 4.85 s, 90% preserved. Medium (5): 4.47 s, 75% preserved. Large (3): 4.31 s, 65% preserved (more coordinates change in large refactorings).

The analysis time scales linearly with the number of coordinates, as predicted by Proposition 4.1. The dominant cost is the initial site construction phase, which processes the full AST of both versions. The functor construction and descent checking phases are sub-linear due to hash-based coordinate lookup.

Morphism structure. The morphism distribution across the benchmark is: inclusion morphisms 12 (8.2%), restriction morphisms 91 (62.3%), and transport morphisms 43 (29.5%), totaling 146 morphisms. Restriction morphisms dominate because function-to-function dependencies (call edges) are the most common relationship in Python code.

Descent strategy comparison. All four strategies were tested: Eager (100.0%, 0.0 ms); Exhaustive (100.0%, 0.0 ms); Iterative (100.0%, 0.0 ms); Optimistic (100.0%, 0.0 ms). Eager is recommended for migration verification due to its fast feedback.

Equivalence checking. For preserved coordinates, 8 equivalence pairs were checked with 8 both verified, 8 same structure, in mean time 11.06 s.

10.10 Case Study: Framework Migration

To illustrate the migration framework on a realistic scenario, we consider migrating a web application from a synchronous HTTP framework to an asynchronous one (analogous to migrating from Flask to FastAPI in the Python ecosystem).

The migration involves the following changes to the semantic site:

- *Handler coordinates* change signatures: synchronous functions become `async def` coroutines. This is a signature-altering change that preserves coordinate identity but modifies propositions.
- *Middleware coordinates* are restructured: the middleware chain becomes an asynchronous pipeline. This is a structural change introducing new coordinates and merging old ones.
- *Database access coordinates* change from synchronous queries to async session management. Both the signature and the dependency structure change.

The change-of-site functor maps 12 of 15 old coordinates to new ones. Of these, 8 are preserved (renamed but semantically equivalent), 3 are modified (require re-verification due to async semantics), and 1 is obstructed (the middleware merging creates a covering incompatibility). The remaining 3 old coordinates are deleted, and 4 new coordinates are added.

Automated repair via `repair_semantics` resolves the middleware obstruction by finding an alternative covering family compatible with the async pipeline. The total migration analysis time is under 10 seconds, compared to a manual re-verification effort estimated at several hours.

10.11 Threats to Validity

Construct validity. Our simulated migrations may not capture the full complexity of real-world migrations. We mitigate this by applying controlled modifications that mirror common refactoring patterns: renaming, parameter addition, module splitting, and interface changes.

Internal validity. The classification of coordinates as "preserved" relies on semantic equivalence checking, which may have false positives for complex behavioral properties. Our equivalence checker uses structural comparison supplemented by SMT-based semantic checks, but cannot guarantee completeness for all property types.

External validity. The benchmark consists of 18 Python programs of moderate size. Results may differ for larger codebases, other languages, or codebases with extensive metaprogramming or dynamic dispatch. We plan to extend the benchmark to larger programs and additional languages in future work.

Reproducibility. All benchmark programs, migration scripts, and analysis code are included in the JuGeo distribution. Experiments were run on a single machine with results reported as means over three trials.

11 Related Work

Code migration tools. Tools like Google's Rosie [1] perform large-scale code modifications. Our approach differs by providing formal guarantees about which verifications survive migration, rather than just syntactic transformation. Facebook's Codemod [9] automates large-scale refactoring but similarly lacks verification transfer. More recently, Comby [10] uses structural matching for language-agnostic code transformation, but does not model the semantic implications of changes.

Proof repair. Ringer et al. [2] study how to repair formal proofs after program changes. Our change-of-site functor provides a principled framework for understanding which proofs need repair and which transfer directly. Their Pumpkin Patch tool [11] searches for proof patches across type equivalences, which can be viewed as a special case of our repair morphism framework.

Incremental verification. Leino and Wüstholz [3] study incremental verification in Dafny, rechecking only changed procedures. Our approach is more granular, operating at the level of individual propositions and using the morphism structure to determine the minimal re-verification set. Logozzo and Fähndrich [18] study modular verification for evolving programs, focusing on contract inference rather than migration planning.

Categorical semantics of refactoring. Roberts [4] formalizes refactoring as behavior-preserving program transformations. Our change-of-site functor generalizes this by modeling not just behavior preservation but judgment transfer across structural changes. Eilenberg and Mac Lane's original category theory work [12] provides the mathematical foundations we build upon.

Semantic versioning formalization. Preston-Werner's Semantic Versioning [13] provides an informal specification for version numbering. Several efforts have formalized semantic versioning using type theory [14]: a version bump is "major" when the API type changes incompatibly. Our change-of-site functor provides a finer-grained analysis: a migration that is "major" in semantic versioning terms may still preserve most judgments if the covering structure is compatible.

Proof refactoring in Coq and Lean. Wibergh [15] studies automated proof refactoring in Coq when definitions change. Ebner et al. [7] describe Lean 4's approach to proof maintenance. Our framework provides a topos-theoretic foundation for understanding when proof refactoring is necessary and when proofs can be transferred directly.

Database schema migration. Schema evolution in databases [16] shares structural similarities with code migration: both involve mapping objects and relationships from an old schema to a new one. Our change-of-site functor is analogous to a schema mapping functor, and our descent preservation theorem corresponds to data integrity preservation under schema evolution.

API evolution analysis. Dig and Johnson [17] study API evolution patterns, classifying changes by their impact on client code. Our framework quantifies this impact through the migration cost and risk metrics, and provides automated repair strategies for breaking changes.

12 Conclusion

We have presented a categorical framework for planning code migrations using change-of-site functors. The MigrationPlan algorithm classifies each coordinate and computes cost and risk metrics. The descent preservation theorem (Theorem 5.1) guarantees judgment transfer when the push-forward has vanishing H^1. The partial preservation theorem (Theorem 5.5) characterizes which coordinates retain verification even when full preservation fails, and the trust attenuation bound (Corollary 5.6) quantifies the worst-case trust degradation. The obstruction classification theorem (Theorem 5.7) provides a three-level taxonomy of migration failures, and the repair completeness theorem (Theorem 7.2) guarantees that local obstructions can always be resolved automatically.

Evaluation on 18 programs shows 70% of coordinates preserved with zero cost, and automated repair resolves 75% of obstructions. The chain composition theorem (Theorem 6.2) and the monotonicity result (Proposition 6.4) provide a principled basis for multi-step migration planning.

Future work. We identify several directions for future work:

- *CI/CD integration*: embedding migration analysis into continuous integration pipelines, providing automatic migration plans on pull requests.
- *Cross-language migration*: extending the framework to migrations between different programming languages, where the change-of-site functor maps between sites with different underlying categories.
- *Machine learning for coordinate matching*: using learned embeddings to improve the coordinate alignment heuristics, particularly for large-scale refactorings.
- *Interactive migration planning*: providing developer-facing tools that visualize the migration plan and allow interactive resolution of obstructions.
- *Larger benchmarks*: extending the evaluation to hundreds of programs across multiple languages and migration patterns.

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